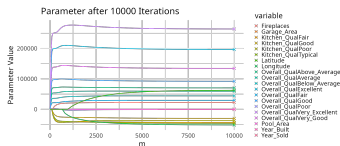
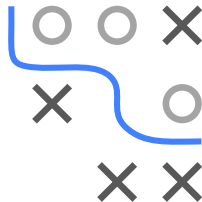


# Introduction to Machine Learning

## Boosting

## Gradient Boosting: CWB and GLMs



### Learning goals

- Understand relationship of CWB and GLM

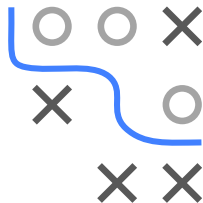
# RELATION TO GLM I

In the simplest case we use linear models (without intercept) on single features as base learners:

$$b_j(x_j, \theta) = \theta x_j \quad \text{for } j = 1, 2, \dots, p \quad \text{and with } b_j \in \mathcal{B}_j = \{\theta x_j \mid \theta \in \mathbb{R}\}.$$

This definition will result in an ordinary **linear regression** model.

- In the limit, boosting algorithm will converge to the maximum likelihood solution.
- By specifying loss as NLL of exponential family distribution with an appropriate link function, CWB is equivalent to (regularized) **GLM**.

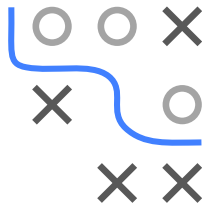


## RELATION TO GLM II

But: We do not *require* an exponential family distribution and we can - in principle - apply it to any differentiable loss

Usually we do not let the boosting model converge fully, but use **early stopping** for the sake of regularization and feature selection.

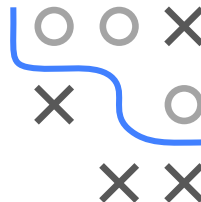
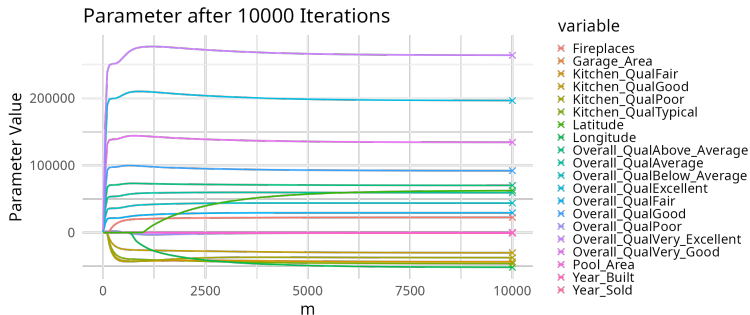
Even though resulting model looks like a GLM, we do not have valid standard errors for our coefficients, so cannot provide confidence or prediction intervals or perform tests etc. → post-selection inference.





# EXAMPLE: CWB PARAMETER CONVERGENCE I

The following figure shows the parameter values for  $m \leq 10000$  iterations as well as the estimates from a linear model as crosses (GLM with normally distributed errors):



Throughout the fitting of CWB, the parameters estimated converge to the GLM solution. The used data set is Ames Housing.